Blockchain Intern Technical Assessment

Carbon Credit Generation and Tokenized Marketplace for EclimAi Ltd

1. Background

EclimAi Ltd helps commercial buildings reduce energy consumption through intelligent monitoring, optimization, and control of building systems.

These energy savings may result in measurable reductions in greenhouse-gas emissions. Subject to an accepted carbon-accounting methodology, independent verification, carbon-registry requirements, and applicable regulations, these emission reductions may potentially qualify for carbon credits.

EclimAi Ltd is exploring whether it can build a platform that:

- Receives energy-savings data from the EclimAi dashboard.
- Converts energy savings into estimated avoided carbon emissions.
- Supports verification and carbon-credit issuance.
- Creates a blockchain-based representation of issued carbon credits.
- Allows approved buyers and sellers to list, purchase, transfer, and retire carbon credits.

Your task is to research this opportunity, recommend a practical approach, and build a limited proof-of-concept prototype within two weeks.

2. Your Objective

You must determine how energy savings recorded by EclimAi Ltd could potentially be converted into carbon credits and represented through a blockchain-enabled marketplace.

Your work should explain the following lifecycle:

1. A commercial building establishes an energy-consumption baseline.
2. EclimAi Ltd measures actual energy consumption after deployment.
3. The platform calculates the energy saved.
4. The energy savings are converted into estimated avoided emissions.
5. The emission-reduction claim is reviewed and verified.
6. Eligible carbon credits are issued through a carbon registry or simulated registry.
7. A blockchain token represents the issued carbon credit.
8. The token is listed, purchased, transferred, and retired.

You must clearly distinguish between:

- Energy savings.
- Estimated avoided emissions.
- Verified emission reductions.
- Officially issued carbon credits.
- Blockchain tokens representing carbon credits.

You must not assume that creating a blockchain token automatically creates a valid carbon credit.

3. Assessment Scope

Your submission must address the following four areas.

3.1 Converting Energy Savings into Carbon Credits

Explain how EclimAi Ltd could convert energy savings from a commercial building into estimated greenhouse-gas emission reductions.

You should cover:

- How a building's baseline energy consumption should be established.
- How actual energy consumption should be measured.
- How energy savings should be calculated.
- How saved electricity or fuel should be converted into tonnes of carbon dioxide equivalent.
- Which emission factors should be used.
- How regional electricity-grid differences should be handled.
- How weather, occupancy, operating hours, seasonal changes, and equipment changes should be considered.
- How additionality should be demonstrated.
- How ownership of the emission reductions should be determined.
- How duplicate claims and double counting should be prevented.

- Which carbon standards, registries, or methodologies may be relevant.
- Whether EclimAi Ltd can issue credits directly or must work with a project developer, independent verifier, carbon standard, or registry.

Include one sample calculation demonstrating how a stated amount of energy savings could be converted into estimated tonnes of carbon dioxide equivalent using an appropriate emission factor.

Clearly state that the result remains an estimate until it is independently verified and accepted by an appropriate carbon-credit standard or registry.

3.2 Carbon-Market Research

Research how existing carbon-credit markets operate.

Your research should include:

- Voluntary carbon markets.
- Compliance carbon markets.
- Carbon-credit standards and registries.
- Carbon exchanges and marketplaces.
- Tokenized carbon-credit platforms.

Select three to five relevant examples and explain:

- What is traded.
- How the credits are created and issued.
- Who the buyers and sellers are.
- How prices are determined.
- Whether trading occurs through an exchange, marketplace, broker, or over-the-counter transaction.
- Whether market makers or liquidity providers participate.
- How ownership is transferred.
- How credits are retired.
- How duplicate claims and double counting are prevented.
- Whether retail users are allowed to participate.
- Whether the platform or market is regulated.

You should also explain the difference between:

- Carbon allowances and carbon credits.
- Primary issuance and secondary trading.
- Spot transactions and forward or futures contracts.
- Holding, transferring, reselling, and retiring a credit.

3.3 Buyer and Seller Analysis

Identify the likely buyers and sellers in an EclimAi carbon-credit marketplace.

Potential sellers may include:

- Commercial-building owners.
- Property-management companies.
- Real-estate investment trusts.
- Energy-service companies.
- Carbon-project developers.
- Aggregators managing multiple buildings.
- EclimAi Ltd or an EclimAi-managed project entity.

Potential buyers may include:

- Companies with net-zero, climate, or sustainability commitments.
- Small and medium-sized businesses.
- Carbon funds.
- Brokers.
- Financial institutions.
- Commodity traders.
- Regulated entities, where the relevant credits are eligible.
- Individuals, where legally and commercially appropriate.

For the most relevant participant groups, explain:

- Why would they buy or sell carbon credits?
- Whether they are buying for retirement, resale, investment, contribution, or compliance.
- Their expected transaction size.
- Their sensitivity to credit quality and price.
- Their due-diligence requirements.
- Their identity-verification, anti-money-laundering, and compliance requirements.

Recommend the most realistic initial buyer and seller groups for EclimAi Ltd.

3.4 Proposed EclimAi Carbon Marketplace

Design a proposed workflow connecting the EclimAi dashboard to a carbon-credit marketplace.

Your workflow should include:

1. A commercial building is registered.

2. Its energy baseline is recorded.
3. Energy-consumption data is received from the EclimAi dashboard.
4. The platform calculates energy savings.
5. The platform converts the energy savings into estimated avoided emissions.
6. Supporting evidence is prepared.
7. The project or emission-reduction claim is reviewed and verified.
8. Eligible carbon credits are issued through a registry or simulated registry.
9. A blockchain token is created to represent the issued credit.
10. The credit is listed for sale.
11. A buyer purchases the credit.
12. Ownership is transferred.
13. The buyer may hold, resell, transfer, or retire the credit.
14. A retired credit becomes permanently non-tradable.

Clearly identify which steps can be performed by EclimAi Ltd and which require an external verifier, carbon standard, registry, legal adviser, financial institution, or regulated service provider.

4. Prototype Requirements

You must build a limited proof of concept that demonstrates the complete workflow.

The prototype does not need to generate officially recognized carbon credits. You may simulate verification, registry approval, payment, settlement, and user onboarding.

You should prioritize functional correctness, auditability, and a clear demonstration over a highly polished user interface.

4.1 Minimum Features

A. Sample EclimAi Data Input

Create a mock API, CSV import, sample dataset, webhook, or manual input form representing data from the EclimAi dashboard.

The data should include:

- Building identifier.
- Building location.
- Reporting period.
- Baseline energy consumption.
- Actual energy consumption.
- Energy saved.

- Energy source.
- Applicable emission factor.
- Estimated avoided emissions.
- Data or verification status.

B. Carbon Calculation

Implement a simple calculation that converts energy savings into estimated tonnes of carbon dioxide equivalent.

Display:

- Formula.
- Input values.
- Units.
- Emission factor.
- Assumptions.
- Calculation result.
- Calculation timestamp or version.

C. Approval Workflow

Implement a basic approval process using statuses such as:

- Draft.
- Data received.
- Calculation completed.
- Pending review.
- Approved.
- Rejected.
- Eligible for issuance.
- Issued.
- Listed.
- Sold.
- Retired.

You may use an administrator account to simulate the verifier or registry.

D. Tokenization

Create a blockchain token representing an approved carbon-credit batch.

The token should contain or reference:

- Project identifier.
- Building identifier.

- Reporting or vintage period.
- Quantity of carbon dioxide equivalent.
- Verification status.
- Registry or simulated registry reference.
- Evidence or source-data hash.
- Current owner.
- Retirement status.

You must briefly justify whether you use:

- Fungible tokens.
- Non-fungible tokens.
- Semi-fungible tokens.
- Individual one-tonne tokens.
- Batch-based tokens.

Explain why your chosen token structure is appropriate for the prototype.

E. Marketplace

The marketplace must allow an approved seller to:

- View issued credits.
- List a credit for sale.
- Set a price.
- Cancel an active listing.
- View completed transactions.

The marketplace must allow an approved buyer to:

- Browse available credits.
- View project and credit details.
- Purchase a credit.
- View owned credits.
- Transfer a credit, where permitted.
- Retire a credit.

F. Retirement

After a credit is retired:

- It must no longer be available for sale.
- It must not be transferable.
- The retirement transaction must be recorded.
- The retiring user must be identified.
- The purpose or beneficiary of the retirement should be recorded.

- The same credit must not be retired twice.
- A retirement record or certificate should be available.

5. Technical Expectations

Your proposal should include a simple system architecture covering:

- Frontend application.
- Backend services.
- Database.
- EclimAi dashboard integration.
- Carbon-calculation service.
- Approval and verification workflow.
- Smart contract.
- Blockchain network.
- Marketplace.
- Evidence storage.
- User accounts or wallets.

You may use a local blockchain or test network.

Acceptable technologies include:

- React, Next.js, Vue, or an equivalent frontend framework.
- Node.js, Python, Java, Go, or an equivalent backend.
- PostgreSQL, MongoDB, SQLite, or an equivalent database.
- Solidity smart contracts.
- Hardhat, Foundry, or an equivalent blockchain development framework.

You must explain:

- Why blockchain is useful for the proposed system.
- Which parts of the system do not require blockchain.
- Which information should remain off-chain.
- Which information should be written or referenced on-chain.
- How confidential building data should be protected.
- How the platform prevents duplicate issuance.
- How the platform prevents double selling and double retirement.
- How the blockchain record would remain aligned with the external carbon registry.

Do not store confidential building telemetry directly on a public blockchain.

6. Smart-Contract Requirements

Your smart contract should demonstrate:

- Authorized credit issuance.
- Unique project or credit identifiers.
- Ownership tracking.
- Transfer of credits.
- Marketplace sale or settlement.
- Prevention of duplicate issuance.
- Permanent retirement.
- Prevention of transfer after retirement.
- Relevant blockchain events for audit purposes.

Include automated tests for at least the following cases:

1. An unauthorized user cannot issue a credit.
2. The same source credit cannot be issued twice.
3. A seller cannot sell a credit they do not own.
4. Ownership changes after a successful purchase.
5. A retired credit cannot be transferred or sold.
6. A credit cannot be retired twice.

7. Security and Compliance

Identify the most important risks associated with your proposed system.

At minimum, discuss:

- Manipulated energy data.
- Incorrect baselines.
- Incorrect emission factors.
- Duplicate project registration.
- Duplicate carbon-credit issuance.
- Double counting.
- Unauthorized token creation.
- Smart-contract vulnerabilities.
- Wallet compromise.
- Loss of private keys.
- Fraudulent verification.
- Registry inconsistencies.
- Misleading environmental claims.
- Privacy and confidentiality.
- Regulatory classification of the token.
- Anti-money-laundering and sanctions requirements.

Recommend practical controls for reducing these risks.

You are not expected to provide formal legal advice. However, you must identify which areas require specialist carbon-market, legal, financial, tax, or regulatory review.

8. Required Deliverables

You must submit the following three deliverables.

Deliverable 1: Written Proposal

Prepare a formal proposal in Google Docs.

Recommended length: approximately six pages, excluding references and appendices.

Your proposal should include:

1. Executive summary.
2. Energy-savings-to-carbon-credit methodology.
3. Carbon-market research.
4. Buyer and seller analysis.
5. Proposed EclimAi marketplace workflow.
6. Technical architecture.
7. Blockchain and token design.
8. Security and compliance considerations.
9. Business-model recommendation.
10. Risks, assumptions, and limitations.
11. References.

Your proposal should be concise. Focus on your recommendations, supporting evidence, and reasoning rather than providing a general introduction to blockchain or carbon markets.

All external information must be properly cited.

Deliverable 2: Prototype and Demo

Build a functional proof of concept and prepare a five-to-ten-minute demonstration.

Your demo should show:

1. A building being registered.
2. Energy data being entered or received from a simulated EclimAi dashboard.
3. Energy savings being calculated.

4. Avoided emissions being calculated.
5. A project being reviewed and approved.
6. A carbon-credit token being issued.
7. The credit being listed for sale.
8. A buyer purchasing the credit.
9. Ownership changing.
10. The buyer retiring the credit.
11. The retired credit becoming non-tradable.

You must also submit:

- Source-code repository.
- Setup instructions.
- Environment-variable template.
- Architecture diagram.
- Smart-contract tests.
- Sample data.
- Known limitations.
- Demo video link or availability for a live demonstration.

Do not include private keys, passwords, credentials, access tokens, or production secrets in the repository.

Deliverable 3: Presentation

Prepare a concise presentation of approximately 8–10 slides.

Your presentation should cover:

- The business problem.
- How EclimAi energy savings could become carbon credits.
- Existing carbon markets and competitors.
- Target buyers and sellers.
- Proposed marketplace workflow.
- Technical architecture.
- Blockchain and token design.
- Prototype demonstration.
- Major security and regulatory risks.
- Final recommendation.

You should be prepared to present your work and answer technical, commercial, and strategic questions.

9. Evaluation Criteria

Your submission will be evaluated out of 100 points.

Carbon-Market Understanding — 20 Points

We will evaluate:

- Your understanding of the carbon-credit lifecycle.
- The accuracy of your energy-to-emissions methodology.
- Your understanding of verification, issuance, ownership, and retirement.
- Your ability to distinguish estimated reductions from issued carbon credits.

Market and Commercial Analysis — 15 Points

We will evaluate:

- The quality of your market research.
- Your understanding of buyers, sellers, exchanges, brokers, and registries.
- The practicality of your recommended initial market.
- The credibility of your proposed business model.

Architecture and Blockchain Design — 20 Points

We will evaluate:

- Your proposed architecture.
- Your token design.
- Your on-chain and off-chain data model.
- Your understanding of blockchain's appropriate role.
- Your approach to registry integration and data integrity.

Prototype Functionality — 25 Points

We will evaluate:

- Completion of the end-to-end prototype workflow.
- Carbon calculation.
- Credit issuance.
- Marketplace listing and purchase.
- Ownership transfer.
- Credit retirement.

- Reliability of the demonstration.

Code Quality and Testing — 10 Points

We will evaluate:

- Code organization.
- Readability.
- Documentation.
- Error handling.
- Smart-contract tests.
- Reproducibility.

Security, Compliance, and Communication — 10 Points

We will evaluate:

- Identification of major risks.
- Controls against fraud and double counting.
- Regulatory awareness.
- Clarity of the proposal, presentation, and demonstration.

10. Submission Requirements

You must submit:

- Google Docs proposal link.
- Presentation link or file.
- Source-code repository link.
- Prototype deployment link, if available.
- Five-to-ten-minute demo video or live-demo availability.
- Architecture diagram.
- Setup and testing instructions.
- List of assumptions and known limitations.
- Reference list.

Submission deadline:

26th July 11:59 PM PST

Make sure all document, presentation, repository, and demonstration permissions are correctly configured before submission.